Trade weighted tariffs

July 22, 2023

This document outlines the steps required to construct the trade weighted GTAP tariffs. I aggregate the GTAP table from 141 countries and 65 sectors to 83 countries and 47 sectors. The structure of this paper is as followed: an analytical representation of the steps required to construct the trade weights followed by a step-by-step example using a simplified 3 country, 3 sector version.

1 Analytical representation

Let's first consider the analytical representation that gets mapped to the data. Our original data consists of a set of importer countries, exporter countries, and exporter sectors denoted by $n_1 \in \mathcal{N}_1$,$i_1 \in \mathcal{I}_1$, and $j_1 \in \mathcal{J}_1$ respectively. Our goal is a to construct a mapping, $T_{\mathcal{N}\times\mathcal{I}\times\mathcal{J}}$, from our original set of of countries and sectors to our new, collapsed version: $T_{\mathcal{N}\times\mathcal{I}\times\mathcal{J}} : \mathcal{N}_1 \times \mathcal{I}_1 \times \mathcal{J}_1 \rightarrow \mathcal{N}_2 \times \mathcal{I}_2 \times \mathcal{J}_2$

For each importer country, n_1, they import some amount $M^{j_1}_{n_1 i_1}$ from country i_1 in sector j_1. Our first task is to consider the total sum of imports for countries and sectors that get mapped to their corresponding aggregation. This is given by for all $n_1 \times i_1 \times j_1 \in T^{-1}(\mathcal{N}_1 \times \mathcal{I}_1 \times \mathcal{J}_1)$, $\sum M^{j_1}_{n_1 i_1} = M'^{j_2}_{n_2 i_2}$. For example, if $T(\{(\text{USA, CAN, coal})\}) = \{(\text{USA, CAN, coal})\}$, then total imports will be the same from before and after the aggregation. However, if $T(\{(\text{USA, ARG, coal}), (\text{USA, ROW, coal})\}) = \{(\text{USA, ROW, coal})\}$ then new total imports would equal the sum of US imports from ARG and ROW in the coal sector, i.e., $M'^{\text{coal}}_{\text{USA,ROW}} = M^{\text{coal}}_{\text{USA,ROW}} + M^{\text{coal}}_{\text{USA,ARG}}$.

To construct the trade weights, we merely divide the original level of imports by the new level of total, aggregated imports: $\gamma^{j_1}_{n_1 i_1} = \frac{M^{j_1}_{n_1 i_1}}{M'^{j_2}_{n_2 i_2}}$. Finally, we multiply these trade weights by the original tariffs and sum according to our aggregating function as before $\tau'^{j_2}_{n_2 i_2} = \sum \tau^{j_1}_{n_1 i_1} \gamma^{j_1}_{n_1 i_1}$.

2 Example

Let's move on to show how this would appear in the data using a simplified example going from 3 countries and 3 sectors to 2 countries and 2 sectors. The table below represents what the original data would look like.

n	i_1	j	$M_{ni_1}^{j_1}$	$\tau_{ni_1}^{j_1}$
USA	ARG	1	200	15
USA	ARG	2	300	0
USA	ARG	3	100	20
USA	ROW	1	500	8
USA	ROW	2	400	4
USA	ROW	3	200	10

We want to collapse ARG into the ROW and sectors 2 and 3 as only sector 2. To construct the appropriate weights for the importing country by sector, I first merge in the corresponding mapping.

n	i_1	j_1	$M_{ni_1}^{j_1}$	$\tau_{ni_1}^{j_1}$	i_2	j_2
USA	ARG	1	200	15	ROW	1
USA	ARG	2	300	0	ROW	2
USA	ARG	3	100	20	ROW	2
USA	ROW	1	500	8	ROW	1
USA	ROW	2	400	4	ROW	2
USA	ROW	3	200	10	ROW	2

I then consider country n's total imports from country i_1 in sector j_1 that map to i_2 in sector j_2.

n	i_1	j_1	$M_{ni_1}^{j_1}$	$\tau_{ni_1}^{j_1}$	i_2	j_2	$\sum_{j_1}\sum_{i_1} M_{ni_1}^{j_1} = M_{ni_2}^{j_2}$
USA	ARG	1	200	15	ROW	1	700
USA	ARG	2	300	0	ROW	2	1000
USA	ARG	3	100	20	ROW	2	1000
USA	ROW	1	500	8	ROW	1	700
USA	ROW	2	400	4	ROW	2	1000
USA	ROW	3	200	10	ROW	2	1000

The trade weights are then given by $\gamma_{ni_1}^{j_1} = M_{ni_1}^{j_1}/M_{ni_2}^{j_2}$

n	i_1	j_1	$M_{ni_1}^{j_1}$	$\tau_{ni_1}^{j_1}$	i_2	j_2	$\sum_{j_1}\sum_{i_1} M_{ni_1}^{j_1} = M_{ni_2}^{j_2}$	$\gamma_{ni_1}^{j_1}$
USA	ARG	1	200	15	ROW	1	700	$\frac{2}{7}$
USA	ARG	2	300	0	ROW	2	1000	$\frac{3}{10}$
USA	ARG	3	100	20	ROW	2	1000	$\frac{1}{10}$
USA	ROW	1	500	8	ROW	1	700	$\frac{5}{7}$
USA	ROW	2	400	4	ROW	2	1000	$\frac{4}{10}$
USA	ROW	3	200	10	ROW	2	1000	$\frac{2}{10}$

Finally, I multiply the original tariff rates by the trade weights and group by i_2 and j_2 to get the new, trade weighted tariffs, $\tau_{ni_2}^{j_2} = \sum_{i_1 \to i_2, j_1 \to j_2} \tau_{ni_1}^{j_1} \gamma_{ni_1}^{j_1}$

n	i_1	j_1	$M_{ni_1}^{j_1}$	$\tau_{ni_1}^{j_1}$	i_2	j_2	$\sum_{j_1}\sum_{i_1} M_{ni_1}^{j_1} = M_{ni_2}^{j_2}$	$\gamma_{ni_1}^{j_1}$	$\tau_{ni_2}^{j_2}$
USA	ARG	1	200	15	ROW	1	700	$\frac{2}{7}$	10
USA	ARG	2	300	0	ROW	2	1000	$\frac{3}{10}$	5.6
USA	ARG	3	100	20	ROW	2	1000	$\frac{1}{10}$	5.6
USA	ROW	1	500	8	ROW	1	700	$\frac{5}{7}$	10
USA	ROW	2	400	4	ROW	2	1000	$\frac{4}{10}$	5.6
USA	ROW	3	200	10	ROW	2	1000	$\frac{2}{10}$	5.6

Our final result is thus given by

n	i_2	j_2	$M_{ni_2}^{j_2}$	$\tau_{ni_2}^{j_2}$
USA	ROW	1	700	10
USA	ROW	2	1000	5.6

The example below show how the operation would work when ARG and ROW are importers.

n_1	i_1	j_1	$M_{ni_1}^{j_1}$	$\tau_{ni_1}^{j_1}$
ARG	USA	1	300	40
ARG	USA	2	150	25
ARG	USA	3	50	50
ARG	ROW	1	400	5
ARG	ROW	2	200	10
ARG	ROW	3	800	40
ROW	USA	1	200	10
ROW	USA	2	500	20
ROW	USA	3	300	30
ROW	ARG	1	100	40
ROW	ARG	2	300	50
ROW	ARG	3	600	60

n_1	i_1	j_1	$M_{ni_1}^{j_1}$	$\tau_{ni_1}^{j_1}$	n_2	i_2	j_2	$M_{n_2i_2}^{j_2}$
ARG	USA	1	300	40	ROW	USA	1	500
ARG	USA	2	150	25	ROW	USA	2	1000
ARG	USA	3	50	50	ROW	USA	2	1000
ARG	ROW	1	400	5	ROW	ROW	1	500
ARG	ROW	2	200	10	ROW	ROW	2	1500
ARG	ROW	3	800	40	ROW	ROW	2	1500
ROW	USA	1	200	10	ROW	USA	1	500
ROW	USA	2	500	20	ROW	USA	2	1000
ROW	USA	3	300	30	ROW	USA	2	1000
ROW	ARG	1	100	40	ROW	ROW	1	500
ROW	ARG	2	300	50	ROW	ROW	2	1500
ROW	ARG	3	600	60	ROW	ROW	2	1500

n_1	i_1	j_1	$M^{j_1}_{ni_1}$	$\tau^{j_1}_{ni_1}$	n_2	i_2	j_2	$M^{j_2}_{n_2i_2}$	$\gamma^{j_1}_{ni_1}$	$\tau^{j_2}_{n_2i_2}$
ARG	USA	1	300	40	ROW	USA	1	500	$\frac{3}{5}$	28
ARG	USA	2	150	25	ROW	USA	2	1000	$\frac{15}{100}$	25.25
ARG	USA	3	50	50	ROW	USA	2	1000	$\frac{5}{100}$	25.25
ARG	ROW	1	400	5	ROW	ROW	1	500	$\frac{4}{5}$	12
ARG	ROW	2	200	10	ROW	ROW	2	1500	$\frac{2}{15}$	56.7
ARG	ROW	3	800	40	ROW	ROW	2	1500	$\frac{8}{15}$	56.7
ROW	USA	1	200	10	ROW	USA	1	500	$\frac{2}{5}$	28
ROW	USA	2	500	20	ROW	USA	2	1000	$\frac{50}{100}$	25.25
ROW	USA	3	300	30	ROW	USA	2	1000	$\frac{30}{100}$	25.25
ROW	ARG	1	100	40	ROW	ROW	1	500	$\frac{1}{5}$	12
ROW	ARG	2	300	50	ROW	ROW	2	1500	$\frac{3}{15}$	56.7
ROW	ARG	3	600	60	ROW	ROW	2	1500	$\frac{6}{15}$	56.7

We don't need ROW to ROW tariffs, so after collapsing, our finall table become

n_2	i_2	j_2	$M^{j_2}_{n_2i_2}$	$\tau^{j_2}_{n_2i_2}$
ROW	USA	1	500	28
ROW	USA	2	1000	25.25
ROW	ROW	1	500	0
ROW	ROW	2	1500	0